Herded or Anchored? Testing Conformity in Machine Learning on a Decade of ICLR Submissions

by

Minghan Li

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Supervisor: A/Prof Jianzhong Qi

in the
School of Computing and Information Systems
Faculty of Engineering and Information Technology
THE UNIVERSITY OF MELBOURNE

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Abstract

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Faculty of Engineering and Information Technology

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Machine learning is often said to herd: researchers chase what is popular, and peer review rewards them for it. But a field can converge without copying. Researchers may instead stay close to a record: their own earlier work, or a topic they already know. We call these two routes to conformity *imitation* and *continuity*, and test the herding account where they make different predictions. The data are 43,480 ICLR submissions from 2017 to 2026, including rejected and withdrawn papers; hypotheses were registered before any outcome was examined and re-tested on 11,099 held-out papers. The field did converge on foundation models, but both routes predict this, and the number of topics that appear to be emerging depends on hindsight: between 8 and 29, depending on which years the topic model is fitted to. Where the two routes make different predictions, we find no evidence of herding. Researchers new to a topic propose methods more often than incumbents do, the reverse of our registered prediction. At review, acceptance shows no detectable difference for papers in crowded topics, whereas papers far from their authors' earlier work are accepted less often (a post hoc finding, confirmed on held-out papers). The conformity visible at ICLR is anchored to researchers' own records rather than driven by the crowd.

Declaration of Authorship

I, Minghan Li, declare that this report, titled 'Herded or Anchored? Testing Conformity in Machine Learning on a Decade of ICLR Submissions', and the work presented in it are my own. I confirm that:

- the report comprises only my original work towards Science Research Project (SCIE30001) except where indicated in the preface;
- due acknowledgement has been made in the text to all other material used; and
- the report is within the word limit set for it, exclusive of tables, figures and the bibliography.

Signed: _____

Date: _____

Preface

This report is my own work for Science Research Project (SCIE30001), carried out under the supervision of A/Prof Jianzhong Qi. It has not been submitted for any other qualification, and it has not been published or submitted for publication.

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Abbreviations

CI	Confidence interval (95% unless stated)
FM	Foundation model
ICLR	International Conference on Learning Representations
LLM	Large language model
MDE80	Minimum detectable effect at 80% power
ML	Machine learning
NLP	Natural language processing
pp	Percentage points
RQ	Research question

Symbols

$F_{\max}$	closeness to the team's earlier ICLR work	cosine similarity
k	number of topics in a topic map	–
κ	Cohen's κ, agreement between two labelling passes	–

Chapter 1

Introduction

1.1 Motivation

Machine learning (ML) is often said to herd. Between 2017 and 2026, yearly submissions to the International Conference on Learning Representations (ICLR) in our discovery set (Chapter 3) grew 40-fold, from 384 to 15,411. Over the same years, the share of submissions building on a foundation model (FM) rose from 0.3% to 54.4%, while the share proposing a new architecture fell from 24.5% to 7.5%. Individual keywords rose and fell (Figure 1.1): generative adversarial networks declined from 8.1% of submissions in 2018, and large language models (LLMs) reached 16.7% in 2026. Critics of ML

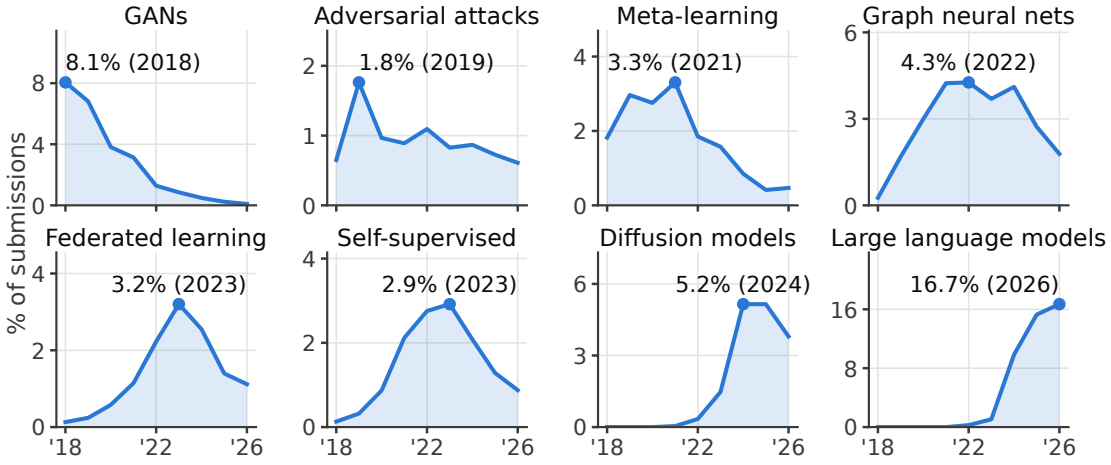


FIGURE 1.1: Keyword trends at ICLR: share of submissions in the discovery set listing each author keyword, 2018–2026; dots mark the peak. The eight keywords were chosen for display after their curves were known. Across the 68 most frequent keywords, 17 peaked in 2019–2024 with shares of at most half the peak both before it and in 2026; Section 4.1 calls these waves.

scholarship point to incentives that favour short-term success [1] and to convergence on whatever suits the field's hardware, benchmarks and foundation models [2–4]. Theories of fashions and information cascades explain such waves by people taking their cue from others [5–7].

A field can converge without copying, however. The switch to FMs, for example, is predicted equally by a bandwagon, by lock-in to a shared tool and by the efficient choice of a tool that works, so the rising FM share alone cannot tell these accounts apart. To decide whether a field herds, one has to look at behaviour: what researchers write when they enter a topic, and what peer review accepts.

This report separates two ways in which researchers can conform. In the first, *imitation*, the reference point is what others are doing now: what is popular, rising or recently accepted. In the second, *continuity*, the reference point is a record: the team's own earlier work, or a topic that a team member already knows. Imitation makes teams alike, whereas continuity keeps each team near its own past. We call work that conforms by continuity *anchored*.[1] The herding account of ML rests on imitation.

1.2 What is missing in the literature

Whether science has fashions is an old question [10], and research about research now studies it with large data [11]. Chapter 2 reviews this work; three gaps motivate the present study.

First, most studies of fashions and herding measure how much an idea is talked about or adopted, not how each adopter is judged. Studies of novelty and its reward look at citations, grants or journals, usually for published work only [12–14]. Few observe in one population both what researchers choose to write and what the gatekeepers accept.

Second, ML venues have been studied mainly for bias and noise in review. Prior work relates ICLR acceptance to prestige, process and research area [15–20], and describes which papers start topics, who enters LLM research and how far researchers move from their own past work [21–23]. None of these studies asks what kind of contribution

[1] 'Anchored' describes a position relative to a record and implies no particular mechanism. It does not refer to the heuristic of anchoring and adjustment [8], the sense in which writing on herding in science uses the word [9].

newcomers to a topic make, or whether acceptance follows the crowd or the team's own record.

Third, histories of a research field usually fit one topic model to the whole archive and read trends forward [24, 25]. Whether the waves found in this way survive when the topics are defined without knowledge of later years has rarely been checked [26, 27].

To our knowledge, no study has tested the herding account both at entry to a topic and at the acceptance decision within one submission population, using hypotheses registered in advance and re-tested on held-out papers. Such a design has been applied to a prestige gap at ICLR [17], but not to herding.

1.3 Research questions

If ML herds, three claims should hold: the waves are fashions, newcomers to a topic carry them by following others, and peer review favours following. Each claim applies at one level of the field, and the report asks one research question (RQ) for each.

RQ1 **The field.** Which changes in ICLR research are passing waves and which persist, and does the answer depend on how topics are measured?

RQ2 **Entry.** When researchers enter a topic, do they follow others, by writing follow-on work that benchmarks or evaluates existing methods, or do they bring work of their own?

RQ3 **The gate.** Is acceptance associated with following the crowd (crowded topics, branded titles, popular keywords) or with staying close to the team's own earlier work?

Imitation and continuity make different predictions for RQ2 and RQ3, but the same prediction for the shift to FMs. Chapter 3 states these predictions as hypotheses.

1.4 Approach

The data are 43,480 ICLR submissions from 2017 to 2026, collected from OpenReview, including rejected and withdrawn papers. Because rejected papers are included, the

study can see what newcomers write for every paper they submit, and can compare accepted with rejected papers of the same topic and year.

Each paper is placed in a topic by clustering text embeddings of its title and abstract, and each is given a contribution type (for example, method or benchmark) by an LLM with a fixed codebook. Researchers' histories are reconstructed from their earlier ICLR submissions, which gives each paper an entry status (newcomer or incumbent in its topic) and a distance from its team's earlier work.

The analysis follows a confirmatory protocol. Hypotheses, models and a rule for when a hypothesis counts as supported were registered before the full corpus was labelled, and acceptance decisions were examined only once, in a single pre-specified run. The main tests were then repeated on 11,099 held-out papers that had not been analysed before. This is a pre-registered out-of-sample replication, not a certified untouched holdout. Each finding carries one of three labels. A REPLICATED finding comes from a registered test whose discovery result, in either direction, was confirmed on held-out papers by a test registered before the held-out data were examined. A HELD-OUT SUPPORTED finding was found post hoc and then supported by a test declared before the held-out data were examined. An EXPLORATORY finding rests on the discovery papers alone or is descriptive. Tests that reach no verdict are called inconclusive or uninformative, as defined in Section 3.4. All results are associations; none shows what would change if a researcher chose a different topic or team.

1.5 Contributions

This report makes three contributions. The first concerns entry (RQ2). It gives a registered test of herding at the point where imitation predicts it most sharply, and finds the reverse of the prediction: researchers entering a topic propose methods more often than incumbents, by 3.7 percentage points (pp), a REPLICATED result. The second concerns the acceptance decision (RQ3). Using all decided submissions under double-blind review, rejected ones included, it finds no evidence that following the crowd is associated with acceptance, while papers far from the team's own earlier work are accepted less often (HELD-OUT SUPPORTED). Hill et al. [28] relate moving away from one's own research to publication success; this report adds the accept or reject decision at an ML venue. The

third contribution is about method (RQ1). The same papers yield between 8 and 29 emerging topics, depending on the years the topic model is fitted to (EXPLORATORY), so refitting the topic model on early years is a cheap check on claims about trends.

1.6 Report outline

Chapter 2 reviews research about research: fashions and herding in science, the reward of novelty, newcomers to research fields, peer review at ML venues, meta-research on ML and NLP, and the measurement of topics over time. Chapter 3 states the hypotheses and describes the data, measures and testing protocol. Chapter 4 reports the results for the field, for entry to topics and for the acceptance decision. Chapter 5 lists the key findings, discusses limitations and suggests future work.

Chapter 2

Literature Review

This chapter reviews research about research that bears on the three research questions. Section 2.1 covers fashions, herding and bandwagons in science and organisations. Section 2.2 covers how science balances tradition and novelty, and how novel work is rewarded. Section 2.3 covers newcomers and researchers who move between topics. Section 2.4 covers peer review at ML venues, Section 2.5 covers meta-research on ML and natural language processing (NLP), and Section 2.6 covers how research topics are measured over time. Section 2.7 summarises what is missing.

2.1 Fashions, herding and bandwagons

The idea that science has fashions goes back at least to Crane [10]. In organisation studies, Abrahamson [7] described management fashions, in which ideas spread because others adopt them rather than because they work, and Abrahamson and Rosenkopf [29] showed that bandwagons can spread even when adopting brings no gain. Abrahamson and Fairchild [30] followed one such fashion over its whole life and found that the way people wrote about it differed between its rise and its fall. Institutional theory gives a related reason for convergence: under uncertainty, organisations copy one another [31], especially organisations that look successful [32]. Strang and Macy [33] modelled firms that imitate their most successful peers and showed that this can produce cycles of adoption and abandonment.

Economic models explain why imitation can be rational for each person and still harmful overall. In an information cascade, people follow those who chose before them and ignore their own information [5, 6]. The resulting consensus can rest on very little evidence and can break easily. Herding also protects a reputation when people are judged against their peers [34]. Diffusion research describes how a new practice spreads through a population over time, from early adopters to late ones [35], and shows that less connected people tend to adopt later [36].

Several authors apply these ideas to science. Fujimura [37] described a bandwagon in cancer research that formed around a package of theory and methods that many laboratories could adopt. Baddeley [9] reviewed herding and social influence in scientific research. In formal models, Weisberg and Muldoon [38] separate scientists who follow others' approaches from those who ignore them, and Weisberg [39] model herding as a way to avoid risk. Alexander et al. [40] dispute some of the conclusions of the first model, and Kitcher [41] and Strevens [42] ask how credit and the priority rule shape how scientists divide their effort. A randomised trial in ML review found no herding among reviewers in discussions [43], while a study of ICLR rebuttals found that reviewers respond to their co-reviewers' scores [44].

There is also a body of work on staying close to one's own past, which this report calls continuity. Organisations learn from their own experience [45] and search near what they already know [46, 47]. Taking up new knowledge depends on related knowledge that one already has [48]. Small early events can lock a population into one technology [49]. These accounts also predict that a field converges, but through each group's own record rather than through copying others.

Two limits of this literature matter here. It mostly measures how much an idea is adopted or discussed, and rarely observes how each adopter is judged. It also rarely separates copying others from staying close to one's own record, although both lead to convergence.

2.2 Tradition, novelty and their reward

The science of science studies research itself with large data [11]. A central theme is the tension between building on a tradition and breaking from it [50]. Foster et al. [12] found

that most biomedical researchers choose strategies that extend existing knowledge, and that more innovative strategies are rarer, riskier and more variable in their payoff. Uzzi et al. [13] found that the most cited papers combine mostly conventional prior work with a small amount of unusual combinations. Wang et al. [51] found that highly novel papers are less cited in the short run and more cited in the long run. At the scale of whole fields, Park et al. [52] report that papers have become less disruptive over time, Wu et al. [53] report that large teams develop existing work while small teams disrupt it, and Chu and Evans [54] report that in very large fields new papers find it harder to become widely cited. Rzhetsky et al. [55] show that choices that are safe for each scientist can make the whole field explore less. Xiang et al. [56] review how researchers choose topics.

Whether evaluators penalise novelty is contested. In grant review, Boudreau et al. [14] found that evaluators gave lower scores to highly novel proposals and to proposals close to their own expertise. In review files from 49 journals, Teplitskiy et al. [57] found no such penalty. Lane et al. [58] showed in field experiments that evaluators lower their scores more after seeing critical peer scores than they raise them after seeing favourable ones. Li [59] found that grant reviewers favour applications in their own area, but are also better informed about them. Cumulative advantage, in which those already recognised gain more recognition, is a further reason why a record can matter at the gate [60]. Priority and timing also matter to scientists [61].

These studies measure reward mostly through citations, grants or published papers. They seldom see, in the same population, both what researchers choose to write and how the gate judges every submission, including rejected ones.

2.3 Newcomers and moving between topics

A long line of work finds that people at the margin of a field bring new ideas. On an open innovation platform, problems were more often solved by people whose expertise lay far from the problem's field [62]. Students from groups under-represented in their field introduce more novel ideas, but these ideas are taken up less [63]. Younger researchers try newer ideas [64], and fresh teams produce more novel work [65]. When an eminent scientist dies, researchers from outside the field enter it and their papers are often highly

cited [66]. Inventors who move into new fields produce more novel but less valuable patents, and the loss is smaller when they work with experts [67].

Moving away from one's own past work has costs. In the pivot penalty, larger shifts from a researcher's earlier work go with lower impact and lower publication success [28]. Researchers who stayed close to their own record during the deep learning shift had lower later impact [23]. Crowding can also push researchers away: after the Soviet collapse, American mathematicians moved out of the fields that received many Soviet mathematicians [68]. Ties to insiders help newcomers: junior scientists who publish with established authors do better later [69], and senior expertise offsets the disadvantage of newcomers to a new research frontier [70]. In ML, half of the first authors of LLM papers on arXiv in 2023 were new to LLM research [22], NLP researchers increasingly publish at ML venues [71], and at ICLR some papers act as catalysts that come before a topic grows [21].

This work measures how novel newcomers' work is, or how they fare later. It does not ask what type of contribution newcomers bring, for example a new method or a benchmark of existing ones, and it rarely looks at the accept or reject decision under double-blind review.

2.4 Peer review at ML venues

Peer review at ML venues is noisy. When the same papers were reviewed by two separate committees, many papers accepted by one were rejected by the other [72, 73]. Shah [74] surveys the problems of review at computer science venues, such as bias, miscalibration and strategic behaviour.

ICLR is a common object of study because its reviews and decisions are public on OpenReview. Kang et al. [75] released a dataset of drafts and reviews from ICLR and other venues and showed that acceptance is partly predictable from the text. Tran et al. [15] and Zhang et al. [76] studied how acceptance at ICLR relates to authors and institutions, and Sun et al. [16] and Manzoor and Shah [77] used ICLR's switch to double-blind review in 2018 to show that blinding reduced, but did not remove, a prestige effect. Where reviewers can see the authors, famous authors fare better [78], and preprints can reveal who wrote a paper even under double-blind review [79]. Ibrahim

et al. [17] tested a prestige gap at ICLR with a pre-registered out-of-sample test. Other studies relate ICLR acceptance to the review process [18] and to research area [19, 20].

These studies ask mostly who gets accepted. They do not ask how acceptance relates to a paper's position in a changing field: whether it is in a crowded topic, whether its authors are new to the topic, or how far it is from the team's earlier work.

2.5 Meta-research on machine learning and NLP

ML has a growing literature that examines its own research practice. Lipton and Steinhardt [1] describe trends in ML writing, such as claims that go beyond the evidence. Several authors argue that ML converges for reasons other than merit: on ideas that suit the available hardware [2], on a few benchmarks [3, 80, 81] and on foundation models that many applications share [4]. Others note the growing role of industry and compute in deep learning [82, 83] and the values that highly cited ML papers express, among them performance, novelty and building on past work [84]. On the other side, Sutton [85] argues that convergence on general methods that scale with computation has been the efficient choice.

Studies of NLP describe paradigm shifts in the field. Interviews and citation analysis show cycles of exploration and exploitation [86, 87], and an analysis of tasks, methods and datasets in NLP papers found that methods and tasks lead and datasets follow [88]. Klinger et al. [89] found that the thematic diversity of AI research has stagnated.

Most of this literature consists of essays, interviews or descriptive trends. Few studies test claims of herding or homogenisation with hypotheses fixed in advance.

2.6 Measuring research topics over time

Computational histories of a field usually fit a topic model to all its papers and then read how topics rise and fall. Griffiths and Steyvers [90] did this for PNAS abstracts, Hall et al. [24] and Anderson et al. [25] for computational linguistics, and Schäfermeier et al. [91] for ML. Dynamic topic models and BERTopic also use all years when fitting

[92, 93]. Prabhakaran et al. [94] showed that the role a topic plays in abstracts, for example as a method or as a task, helps predict whether it will grow.

Judging which topics are emerging is hard, because emergence is partly defined by what happens later [26], and a classification of topics fitted at one time may not suit later years [95]. Li et al. [27] evaluated how well static topic models detect emerging topics, and Khanbayov and Kurban [96] tested a signal for emerging AI topics that uses only earlier data. In prediction, information that would not have been available at the time, known as leakage, makes results look better than they are [97, 98]. Hindsight bias is the same problem in human judgement [99].

This report also uses tools for confirmatory research on existing data. Pre-registration means fixing hypotheses and analyses before seeing the outcomes, and it can be applied to data that already exist [100, 101]. Multiverse and specification curve analyses run a result under many reasonable choices to see how much it depends on them [102, 103], and analyses chosen after looking at the data should be treated as exploratory [104]. Labels produced by LLMs are increasingly used in such research, but their agreement with human coders varies by task [105].

2.7 Summary and research gap

The literature offers two broad accounts of why a field converges: imitation of others, and continuity with one's own record. It provides evidence on each, but mostly in separate settings and for published work. Studies of fashions measure adoption rather than reward. Studies of novelty and newcomers measure reward through citations, grants or journals and rarely see the decision on every submission. Studies of ML review focus on prestige and noise rather than on herding, and histories of ML and NLP fit topics with hindsight.

This report brings these strands together in one population. It asks whether ICLR's waves survive measurement without hindsight (RQ1), what kind of contribution newcomers bring to a topic (RQ2), and whether acceptance follows the crowd or the team's own record (RQ3). It uses hypotheses registered in advance and tests the main ones again on held-out papers.

Chapter 3

Method

This chapter states the hypotheses (Section 3.1), describes the data (Section 3.2) and the measures (Section 3.3), and sets out the testing protocol (Section 3.4). Section 3.5 notes two weaknesses of the design.

3.1 Hypotheses

Table 3.1 sets out what imitation and continuity predict for each research question. Under imitation, researchers entering a topic face the most uncertainty, so they should flock to hot topics, adopt rising tools early and write follow-on work that benchmarks or evaluates what others have proposed. If review rewards following, papers in crowded topics, papers with branded titles (a short coined name and a colon) and papers with keywords near their peak should be accepted more often. Under continuity, incumbents should consolidate a topic's existing body of work, newcomers should bring work from elsewhere, new tools should spread through teams that already have ICLR work, and acceptance should track how close a paper stays to its team's earlier work.

Some rival explanations make the same predictions. Visible status, cumulative advantage and expertise [17, 59, 60] also predict that acceptance is associated with a team's record. Network diffusion, in which less connected people adopt later [36], predicts the same as continuity for new tools, namely that teams new to ICLR take them up later.

Not all of these predictions were registered in advance. Table 3.2 lists the hypotheses that were registered before any outcome was examined. They cover imitation's predictions

Route (reference point)	Field (RQ1)	Entry (RQ2)	Gate (RQ3)
Imitation (others, now)	Convergence on shared tools	Entrants flock to hot topics, adopt rising tools early and write follow-on work that benchmarks or evaluates others' methods	Acceptance responds to crowding (the registered test predicted a penalty); a premium for branded titles and for keywords near their peak
Continuity (own record)	Convergence on shared tools	Entrants bring work from other topics; incumbents consolidate; new tools spread through teams with earlier ICLR work	Acceptance tracks closeness to the team's own record

TABLE 3.1: What each route predicts at the three levels. Lock-in and efficient choice also predict convergence on shared tools, so only entry and the gate can tell the routes apart.

ID	RQ	Prediction	Sign
H1a–c	1	As a topic accumulates papers, its share of method papers falls and its shares of measurement and empirical papers rise	−, +, +
H1d	1	Where more of a topic's papers built on FMs in the previous year, fewer papers propose a new architecture	−
H1e	1	... and more papers are empirical or evaluation studies	+
H2a	2	Hot topics have a lower share of incumbent papers (entrants flock to them)	−
H2b	2	Newcomer papers are less often method papers	−
H2c	2	Newcomer papers are more often measurement papers	+
H3a	3	Papers in more crowded topics are accepted less often	−
H3b	3	This penalty is larger for method papers	−

TABLE 3.2: Hypotheses registered before any paper in the full corpus was labelled. All models include topic and year fixed effects (Section 3.4).

for entry and the crowding test at the gate, plus tests of how a topic's contribution mix changes as it grows. The registered crowding test predicted a penalty for crowded topics, whereas imitation, if review rewards it, predicts a premium; a result of either sign would therefore show review responding to the crowd. Continuity's predictions were written after the tests had been run and are used only to organise the results. Two further hypotheses on the use of fashionable keywords (H2d and H3c) were dropped before any estimate, because their measure failed its validation check.

3.2 Data

The corpus was built from official OpenReview submission records, which were downloaded on documented dates. It contains 55,938 ICLR submissions from 2017 to 2026,

Year	Discovery	Held-out
2017	384	105
2018	770	212
2019	1,248	309
2020	2,070	500
2021	2,360	603
2022	2,649	704
2023	3,872	995
2024	5,765	1,445
2025	8,951	2,284
2026	15,411	3,942
All	43,480	11,099

TABLE 3.3: Papers with a usable abstract by year. 69.3% of discovery papers are from 2024–2026.

including rejected and withdrawn papers. Titles, abstracts, author keywords, author lists and decisions all come from these records.

Before the analysis began, about one fifth of the submissions were selected at random and set aside as a held-out set. The remaining submissions form the *discovery set*: the 43,480 papers with a usable abstract (at least 30 words). The discovery set was used to develop and run the registered tests. The *held-out set* contains 11,099 papers with an abstract; 1,309 submissions whose links to other submissions were unresolved were kept out of both sets. The held-out papers were not analysed until after the discovery tests had been registered and run, and they were then used only for tests declared in advance. Because the two sets share conference years, reviewers and authors, the held-out tests guard against overfitting by the analyst but not against patterns specific to a year or to a pool of reviewers. For this reason the held-out analysis is described as a pre-registered out-of-sample replication, not a certified untouched holdout. Table 3.3 gives the number of papers by year.

Different questions use different subsets. The field analyses use all discovery papers. The entry analyses use the 42,325 papers from 2019–2026 that have at least one author identifier. The acceptance tests use the 32,190 discovery papers from 2018–2026 that were accepted or rejected (40.1% were accepted) and the 8,185 held-out papers from the same years that were accepted or rejected. Papers from 2017 serve only as earlier history, and the 9,986 withdrawn discovery papers are left out of the acceptance tests. The field and entry analyses use no acceptance decisions.

Existing collections of ICLR data were checked but not used as the main source. The

Re^2 dataset [106] was considered first. Its coverage of the main ICLR conference ends in 2023 and omits 2019, so this study collected its own data.

3.3 Measures

3.3.1 Topics

A *topic map* is one clustering of all papers into 100 topics. Each paper's title and abstract are turned into a vector with SPECTER2, a text encoder trained on scientific papers [107, 108], and the vectors are grouped by spherical k-means with $k = 100$. A *topic* is one cluster. A topic map can be fitted on papers from any set of years and then applied to all papers. Unless another map is named, topics come from the map fitted on all years. Alternative maps (other values of k, a map that weights each year equally, and maps fitted on 2017–2020 only) are used as robustness checks.

3.3.2 Contribution types

Each paper is given one primary contribution type by an LLM (Claude Haiku) that follows a fixed codebook. The types are method (including a new architecture), theory, empirical study, dataset or benchmark, evaluation, and other. A *method paper* proposes a method. A *measurement paper* contributes a dataset, benchmark or evaluation. An *FM paper* uses or adapts a large pre-trained model, such as an LLM; this label is coded with the same codebook. Over all discovery papers, 75.1% are method papers and 7.6% are measurement papers.

The labels were checked in two ways. First, the same LLM labelled a sample of 2,174 discovery papers a second time, in an independent run that could not see the first labels. The two runs agreed at Cohen's $\kappa = 0.93$ (97.5% agreement for method papers and 99.3% for measurement papers). Second, the author checked a random sample of 100 labels by hand against the codebook, without help from anyone else. About 80 of the 100 labels were clearly correct. The other 20 or so were borderline cases in which the author could not decide between two types either, and the LLM's label was kept. No label in the sample was judged clearly wrong, but the borderline cases are not counted as agreement.

These checks show that the LLM is consistent with itself. They do not show how often its labels would match those of trained human coders. Ziems et al. [105] found that LLMs labelling social science text without task-specific training often reach only fair agreement with human coders. A second human coder would be needed to measure this for the present labels, and Chapter 5 lists it as future work.

3.3.3 Entry profiles

Entry status is defined relative to a topic and counts ICLR submissions only, with any outcome. In a *newcomer paper*, no author has an earlier ICLR submission in the paper's topic, following the topic-relative definition of Biliotti et al. [70]; otherwise it is an *incumbent paper*, and an author with such a submission is a *topic insider*. A newcomer paper comes from a *first-time team* if no author has any earlier ICLR submission, and from a *returning topic newcomer* if some author has one in another topic. Authors are matched by their OpenReview identifiers, never by name. In 2021 ICLR moved from e-mail-based to profile-based identifiers, and the two are not linked, so some earlier histories are hidden and some incumbents appear as newcomers. This pulls the entry contrasts towards zero.

3.3.4 Two distances

Both distances are measured in the same SPECTER2 vector space, but from different reference points. *Closeness to the team's own record*, $F_{\max}$, follows Bao and Li [23]. For each author with earlier ICLR submissions since 2021 (an *experienced author*), it is the cosine similarity between the paper and the mean of that author's earlier submissions; $F_{\max}$ is the highest value among the paper's experienced authors. Probable earlier versions of the same paper are excluded. *Distance from earlier ICLR work* is one minus the mean cosine similarity to the ten most similar submissions from earlier years, expressed as a percentile within topic and year.

3.3.5 Predictors measured from the past

Every predictor uses only information that was available before the paper's year. *Prior FM adoption* is the FM share of the topic's papers in the previous year. *Crowding* is

the natural log of one plus the number of the topic's earlier papers. A *hot topic* is one in the top fifth of growth before the paper's year.

Abstracts were downloaded after the decisions, so some accepted papers carry their final version. Links to code repositories, which are usually added after acceptance, appear in 17.6% of accepted but only 0.5% of rejected 2023–2026 abstracts, and so leak the outcome [97]. Removing them and recomputing the text vectors leaves the discovery estimates of the distance-based acceptance tests unchanged. This check could not be repeated for held-out papers.

3.4 Testing protocol

The hypotheses, models and a rule for support were registered in dated documents, kept with the analysis code, before the full corpus had been labelled. Unless stated otherwise, the models are linear probability models with topic and year fixed effects, and standard errors are clustered by topic. For the registered tests, acceptance decisions were read only by pre-committed analysis code, in a single logged run [100, 101].

Each registered test was re-run under pre-specified variants: other topic maps and numbers of topics, the map that weights years equally, and subsets of years and of papers. Tests were grouped into families by research question for the Holm correction for multiple testing [109]. A registered hypothesis counts as supported if its estimate has the predicted sign with a Holm-adjusted $p < 0.05$ in its family, and keeps that sign in at least 75% of the counted variants and in both hindsight variants (the map that weights years equally, and the 2017–2020 map applied to 2021–2026 papers).

Analyses that were not registered at the start (post hoc analyses) were declared in writing before they were run. For each question, a one-sided test on the held-out papers was declared before any held-out estimate for that question was computed. A declared held-out test is supported if its one-sided p is below 0.05. Holm-adjusted values across the six secondary held-out tests are reported alongside, but do not set the label. Because the held-out tests are one-sided, a supported held-out test confirms the sign of an association; its size should not be compared with the discovery estimate.

Tests that are not supported are described in one of two ways. A test is *uninformative* when its estimate lies below its minimum detectable effect at 80% power (MDE80): it cannot tell a small association from none. It is *inconclusive* otherwise, or when no MDE80 is available. The evidence labels REPLICATED, HELD-OUT SUPPORTED and EXPLORATORY are defined in Chapter 1.

3.5 Weaknesses of the design

Two weaknesses should be kept in mind when reading the results. First, the standard errors may be too small. In a simulation in which there was no true effect but papers were related across topics, the main entry tests gave a significant result 12.6% and 10.0% of the time at a nominal 5% level, and a wild cluster bootstrap [110] did not correct this. Second, keyword popularity and title style had been explored against acceptance on the discovery papers before the hypotheses were registered. The discovery estimate for titles is therefore not a test, and the keyword-timing tests probe a dimension that had already been seen.

Chapter 4

Results

This chapter reports the results in the order of the research questions: the field (Section 4.1), entry to topics (Section 4.2) and the acceptance decision (Section 4.3). Table 4.1 first gives the outcome of every registered hypothesis. The findings are listed together in Chapter 5.

ID	Prediction	Discovery [95% CI]	Held-out [95% CI]	Outcome
H1a	Method share falls as a topic grows	−0.14 [−1.8, +1.5]	–	not supported
H1b	Measurement share rises	+0.50 [−0.8, +1.8]	–	not supported
H1c	Empirical share rises	−0.31 [−1.3, +0.7]	–	not supported
H1d	Fewer new architectures where FM adoption is high	−7.98 [−12.14, −3.82]	−11.0 [−18.1, −3.9]	REPLICATED
H1e	More empirical or evaluation papers where FM adoption is high	−0.63 [−3.64, +2.38]	–	INCONCLUSIVE
H2a	Entrants flock to hot topics	−0.73 [−2.68, +1.22]	–	INCONCLUSIVE
H2b	Newcomers write fewer method papers	+3.71 [+2.61, +4.80]	+3.60 [+1.48, +5.71]	reversed; REPLICATED
H2c	Newcomers write more measurement papers	−1.70 [−2.46, −0.93]	−2.10 [−3.35, −0.85]	reversed; REPLICATED
H3a	Crowded topics accepted less often	−1.68 [−3.68, +0.32]	−1.51 [−5.20, +2.19]	UNINFORMATIVE
H3b	Larger crowding penalty for method papers	−0.36 [−1.51, +0.79]	–	not supported

TABLE 4.1: Outcomes of the registered hypotheses (Table 3.2). Units are pp: per log unit of earlier papers for H1a–c and H3a–b, and for a change in prior FM adoption from 0 to 100% for H1d–e. For H2b and H2c the registered prediction failed in the opposite direction; the reversed hypotheses were registered before the held-out data were examined and were supported on held-out papers. Held-out tests are one-sided.

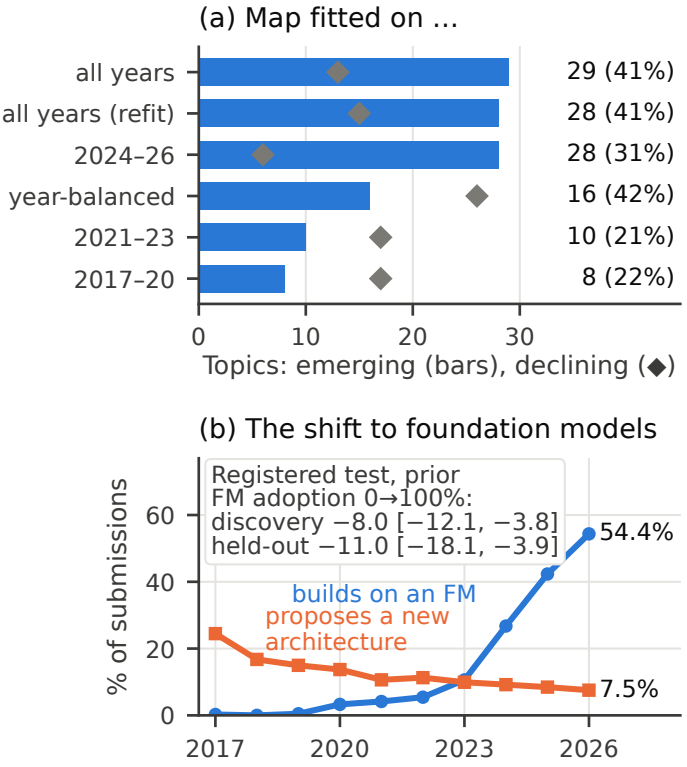


FIGURE 4.1: (a) Emerging (bars) and declining (grey diamonds) topics on topic maps fitted on different years, for the same 43,480 papers. Labels at right show the number of emerging topics and, in brackets, the share of 2024–2026 papers in them (22.3% on the 2017–2020 map, 40.9% on the all-years map); 'all years (refit)' uses another random seed. Topic size 'predicts' two-year growth only on the all-years map (out-of-sample $R^2 = 0.28$ against -0.03; no paired test). (b) Share of submissions building on an FM (blue) and proposing a new architecture (orange), using contribution types assigned by the LLM, with the registered test inset (REPLICATED).

4.1 The field: what is a wave and what lasts (RQ1)

This section asks which changes in the field are passing waves and which persist.

4.1.1 Waves

Author keywords need no topic map, and at their level waves are real: 17 of the 68 most frequent keywords are *waves*, which rose and fell, peaking in 2019–2024 with shares of at most half the peak both before it and in 2026 (Figure 1.1; EXPLORATORY).

4.1.2 Emergence depends on hindsight

A topic is called *emerging* if its 2024–2026 share is at least three times its 2017–2020 share, and *declining* if its 2024–2026 share is at most a third of its 2017–2020 share. With the same papers and pipeline, a map fitted on 2017–2020 finds 8 emerging topics and one fitted on all years finds 29 (Figure 4.1a; EXPLORATORY). Two causes combine, and they cannot be separated. The all-years map uses the future to describe the past, like leakage in prediction [98] and hindsight bias in judgement [99]: with 69% of its papers in 2024–2026, it is in effect a present-day map. The early map, for its part, cannot represent topics born after 2020. The two counts therefore bracket the number of emerging topics.

The registered tests of whether a topic's contribution mix changes as it grows (H1a–c) were not supported on the all-years map (Table 4.1). On the early map, the method and measurement estimates point the predicted way, so whether topics show such a life cycle also depends on how topics are defined.

4.1.3 What lasts: the shift to foundation models

The FM share rose every year from 2018 (Figure 4.1b). Prior FM adoption is associated with fewer new-architecture papers (H1d): from 0 to 100% adoption the difference is −7.98 pp [−12.14, −3.82] in discovery and −11.0 [−18.1, −3.9] on held-out papers (REPLICATED). The association may also run the other way. The test for a parallel shift towards empirical or evaluation papers (H1e) is inconclusive (−0.63 [−3.64, +2.38]). The pattern matches accounts of NLP's paradigm shifts [86–88], but lock-in [2], efficient choice [85] and a bandwagon [37] predict it too. The field level therefore cannot separate the two routes.

4.2 Entry: who writes what (RQ2)

This section asks what researchers write when they enter a topic, and whether they follow others into it as the herding account implies.

Contrast	Discovery [95% CI]	Held-out [95% CI]	Label
Method paper	+3.71 [+2.61, +4.80]	+3.60 [+1.48, +5.71]	REPLICATED
Measurement paper	−1.70 [−2.46, −0.93]	−2.10 [−3.35, −0.85]	REPLICATED
Measurement paper, with log team size[a]	−0.54 [−1.11, +0.04]	−0.98 ($p = 0.10$)	REGISTERED VARIANT
Method: returning topic newcomer − incumbent[b]	+3.43 [+2.31, +4.55]	–	EXPLORATORY
Method: first-time team − returning topic newcomer[b]	+0.92 [−0.44, +2.29]	–	EXPLORATORY
Distance from earlier ICLR work, own work excluded (percentile points)	+4.33 [+3.44, +5.22]	–	EXPLORATORY
Entry into hot topics: incumbent share, hot − other topics[c]	−0.73 [−2.68, +1.22]	–	INCONCLUSIVE

TABLE 4.2: Entry contrasts: newcomer minus incumbent papers unless the row names other profiles, in pp within topic and year. Held-out tests are one-sided. [a]Registered supplementary variant, outside the support rule. [b]First-time team: no author has any earlier ICLR submission; returning topic newcomer: an author has one, in another topic. [c]H2a; no held-out test was registered.

4.2.1 Newcomers propose methods

The registered prediction was that newcomer papers, whose authors face the most uncertainty, would more often be follow-on work: measurement papers that benchmark, collect data for or evaluate what others had proposed (H2b, H2c). Newcomer papers make up 73% of the 42,325 papers in 2019–2026 that have an author identifier. Compared with incumbent papers, the difference in the share of method papers is +3.71 pp [+2.61, +4.80] in discovery and +3.60 [+1.48, +5.71] on held-out papers; for measurement papers it is −1.70 [−2.46, −0.93] and −2.10 [−3.35, −0.85] (Table 4.2, Figure 4.2). Both contrasts are REPLICATED. The registered prediction thus failed in the opposite direction, and the reversed hypotheses, registered before the held-out data were examined, were supported on the held-out papers. All ten counted variants agree in sign, and the t-statistics (6.7 and −4.4) are too large to be explained by the inflated false-positive rate reported in Section 3.5. Team size accounts for much of the measurement gap: with log team size as a control, the gap shrinks to −0.54 [−1.11, +0.04] in discovery and −0.98 ($p = 0.10$) on held-out papers, whereas the method gap barely changes (+3.92 and +3.64).

4.2.2 Topic newcomers and first-time ICLR teams

Newcomer papers split into those from first-time teams (26.3% of papers; median team size 3) and those from returning topic newcomers (46.7%; median 5); incumbent papers make up the remaining 27.0% (median 5). Returning topic newcomers write method papers more often than incumbents (+3.43 pp [+2.31, +4.55]) at the same median team size, and the difference holds with team size controlled (+3.62). For first-time teams the difference is +4.35 [+2.82, +5.89], and the interval for the gap between the two newcomer profiles includes zero (+0.92 [−0.44, +2.29]). All three contrasts are EXPLORATORY. The method surplus is therefore associated mainly with entering a topic, not with being new to ICLR or with working in a small team, and it appears in every period, including before 2023.

The measurement gap behaves differently. For first-time teams it is accounted for by team size (−2.24 [−3.24, −1.25] against incumbents, and +0.09 [−0.63, +0.80] with team size controlled); for returning topic newcomers part of it remains (−1.47, and −0.81 [−1.46, −0.16] with team size controlled). The gap appears only in 2024–2026 (first-time teams against incumbents: −2.75 [−4.07, −1.44]). In the FM era, then, measurement papers are concentrated among topic incumbents and larger teams. Team size is the only resource observed here; others, such as compute, may also matter. NLP shows a parallel at the level of research trends, where tasks and methods lead and datasets follow [88].

4.2.3 Farther from earlier work

Newcomer papers lie farther from earlier ICLR work than incumbent papers. Within topic and year the difference is +5.90 percentile points [+4.95, +6.85]; it is +5.24 among method papers alone and +4.33 [+3.44, +5.22] when the authors' own earlier papers are excluded from the reference set. Papers by returning topic newcomers are less likely than incumbent papers to have a same-year *twin*, an independent submission at cosine similarity ≥ 0.70; with first-time teams controlled separately, the difference in twin share is −3.86 pp [−4.98, −2.74]. These results are EXPLORATORY. The follow-on prediction expects newcomer papers to lie closer to earlier work. Writing method papers could itself be imitation of the venue's usual kind of paper, since three quarters of ICLR papers are

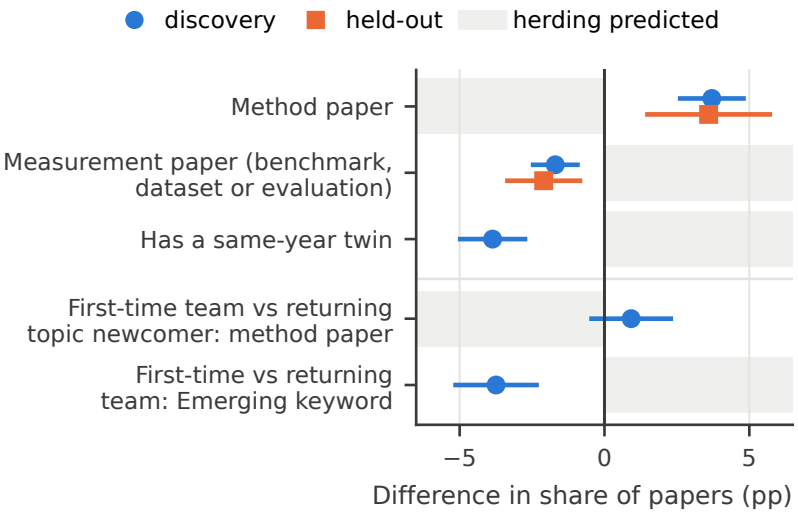


FIGURE 4.2: Entry contrasts within topic and year (pp), with 95% intervals: discovery (blue circles) and held-out (orange squares). Rows 1–2 compare newcomer with incumbent papers; they are registered and REPLICATED. Rows 3–5 are EXPLORATORY (discovery only). Row 3 compares papers by returning topic newcomers with incumbent papers, with first-time teams entered separately and with contribution-type, FM and team-size controls. Row 4 compares first-time teams with returning topic newcomers, and row 5 compares first-time teams with returning teams (all teams with earlier ICLR submissions) on the use of young, rising Emerging keywords. Grey shading marks the side that the herding prediction expected.

method papers, but newcomers' distance from earlier work argues against that reading. Neither contribution type nor embedding distance measures originality; these results argue against newcomers following others but say nothing about whether their work is more original.

4.2.4 Later uptake of foundation models

Among 2023–2026 papers in topics where 5–50% of the previous year's papers built on FMs, the gap in FM-paper share between first-time teams and returning topic newcomers is −4.58 pp [−6.55, −2.60], widening to −14.71 [−19.82, −9.61] as adoption rises. For returning topic newcomers and incumbents the data show no clear difference. First-time teams also list *Emerging* keywords, which are young and rising, at a lower rate than teams with earlier ICLR submissions; the difference is −3.74 [−5.14, −2.35]. Both results are EXPLORATORY. Foundation models thus spread through teams with earlier ICLR work, as absorptive capacity predicts [48]. Network diffusion predicts the same order, however, so this contrast cannot tell the routes apart.

4.2.5 Flocking to hot topics

Whether entrants crowd into hot topics remains open. The registered test (H2a), which compares the incumbent share in hot and other topics, is inconclusive (−0.73 [−2.68, +1.22]), and no held-out test was registered for it. None of the entry contrasts measured here favours the follow-on prediction.

4.3 The gate: crowd or record? (RQ3)

This section asks whether acceptance at ICLR is associated with following the crowd or with staying close to the authors' own earlier work. Table 4.3 and Figure 4.3 collect the tests. Of those in the figure, crowding was registered, title style had been explored before registration and the remaining tests were found post hoc in discovery; each held-out test was declared before any held-out estimate for its question was computed.

4.3.1 Is following the crowd associated with acceptance?

The registered crowding test (H3a) predicted a penalty for crowded topics, and it is uninformative. Per log unit of crowding (the natural log of one plus the number of the topic's earlier papers), acceptance changes by −1.68 pp [−3.68, +0.32] in discovery and by −1.51 [−5.20, +2.19] on held-out papers, against an MDE80 of 4.8. The data show no clear penalty or premium, although held-out premiums above +2.19 pp per log unit lie outside the interval. Whether the penalty is larger for method papers (H3b) could not be told (−0.36 [−1.51, +0.79]).

The clearest sign of imitation in the corpus is in title style. The share of titles that open with a short coined name and a colon rose from 10.9% in 2017 to 44.6% in 2026. Within topic and year, with contribution-type, FM and team-size controls, such branded titles are accepted 1.74 pp [+0.37, +3.11] more often in discovery, but this estimate is not a test, since title style had been explored before registration. The declared held-out test gives +1.76 [−0.82, +4.34] (one-sided $p = 0.089$); the held-out test is uninformative (below its MDE80 of 3.2), so the premium remains EXPLORATORY. Raw within-year gaps are larger (+3.3 in discovery and +4.0 on held-out papers); about half of each

gap is explained by topic, contribution type, FM use and team size. Titles can also be revised after the decision, which could create such a premium.

FM papers are accepted 4.62 pp more often [+2.79, +6.44], with no sign of a difference between late and early adopters: the premium changes by +0.19 [−0.96, +1.34] per logit unit of the topic's prior FM adoption (that is, per unit of its log-odds). Both results are EXPLORATORY, and they fit efficient choice as well as imitation, if review rewards it. A topic's *edge* is its acceptance rate minus that of all other papers of the same year, using topics of the 2017–2020 map. The edge persists but shrinks towards the mean: among discovery papers, the 2018–2022 edge predicts the 2024–2026 edge with slope 0.38 [0.18, 0.58]. The held-out test asks a related but different question, namely whether the discovery edges for 2018–2022 predict the 2024–2026 edges of held-out papers; it gives a slope of 0.60 [0.32, 0.88] and is HELD-OUT SUPPORTED. A favoured topic thus keeps only part of its edge, as expected when decisions carry a large arbitrary component [72, 73].

Keyword timing shows at most weak associations, and it is treated as EXPLORATORY: keyword popularity had been explored against acceptance before registration, and both held-out intervals below include zero. On held-out papers, *Emerging* keywords (young and rising) are associated with higher acceptance than *Mainstream* keywords (old and flat), by +3.25 [−0.18, +6.69], and acceptance may fall after a keyword's share peaks (−5.2 [−11.1, +0.7]; −1.7 without the six keywords examined before this test was declared). Neither pattern shows the constant premium for popular keywords that imitation would predict if review rewarded it.

4.3.2 What is acceptance associated with?

The main pattern is the *far jump*: the lower acceptance of papers far from the team's earlier work. Among 20,430 decided 2023–2026 papers with an experienced author, acceptance at the 10th percentile of closeness to the team's own record is 5.44 pp lower than at the median in discovery (−5.44 [−7.10, −3.78]) and 6.09 pp lower on held-out papers (−6.09 [−9.32, −2.86]), with power = 0.89. This was the primary test declared for the held-out papers, and it is HELD-OUT SUPPORTED. Adjusted acceptance rises with closeness up to the fourth fifth of the distribution. The far jump remains with a newcomer control (−5.0), although closeness is correlated with newcomer status ($r = -0.57$).

Contrast in acceptance	Discovery [95% CI]	Held-out [95% CI]	Label
Referenced to the team's own record			
Far jump: 10th percentile of $F_{\max}$ − median	−5.44 [−7.10, −3.78]	−6.09 [−9.32, −2.86]	HELD-OUT SUPPORTED
Same author, 0.1 closer in cosine similarity[c]	+0.55 [+0.21, +0.90]	+1.06 [+0.23, +1.89]	HELD-OUT SUPPORTED
One topic insider vs none[c]	+3.42 [+1.18, +5.66]	+3.88 [+0.28, +7.47]	HELD-OUT SUPPORTED
Referenced to others or the field			
Crowding, per log unit[a]	−1.68 [−3.68, +0.32]	−1.51 [−5.20, +2.19]	UNINFORMATIVE
Branded title[b]	+1.74 [+0.37, +3.11]	+1.76 [−0.82, +4.34]	EXPLORATORY
Change in FM premium per logit unit of prior adoption	+0.19 [−0.96, +1.34]	–	EXPLORATORY
Topic edge 2018–22 → 2024–26, slope[c]	0.38 [0.18, 0.58]	0.60 [0.32, 0.88]	HELD-OUT SUPPORTED
Emerging − Mainstream keyword[b,c]	+2.67 [+0.76, +4.58]	+3.25 [−0.18, +6.69]	EXPLORATORY
Distance from earlier ICLR work: least-distant fifth − middle fifth	−3.27 [−4.93, −1.60]	−0.13 [−3.50, +3.25]	UNINFORMATIVE

TABLE 4.3: Acceptance contrasts at the gate, in pp within topic and year (topic edges and keywords: against the same year's other papers). Rows differ in population and unit, so their sizes are not comparable. Held-out tests are one-sided. [a]Registered (H3a); predicted a penalty; MDE80 4.8. [b]Explored against acceptance before registration. For titles the discovery estimate is not a test (held-out MDE80 3.2); for keywords it is a declared post hoc test on a dimension already seen. [c]Holm-adjusted p across the six secondary held-out tests: topic-edge slope 0.019, edge prediction 0.026, same-author closeness 0.026, insider 0.052, Emerging keywords 0.063, same-author distance slope 0.22.

Holding the author fixed, a paper 0.1 closer in cosine similarity to that author's earlier work is associated with acceptance 0.55 pp higher [+0.21, +0.90] in discovery and 1.06 pp higher [+0.23, +1.89] on held-out papers. This result is HELD-OUT SUPPORTED and survives a family-wise correction across the secondary tests. Controls for newcomer status and for co-authors' experience change it little (+0.48 and +0.50), though unmeasured differences in co-author quality could also bias it; a simulation run before the held-out test showed that such a confounder can shift the estimate. In a related within-author comparison, on a paper in a topic new to them, the same author's acceptance is 1.37 pp lower (−1.37 [−2.74, −0.004]; author fixed effects, EXPLORATORY).[1]

Having one topic insider on the team, rather than none, is associated with acceptance 3.42 pp higher [+1.18, +5.66] in discovery and 3.88 pp higher [+0.28, +7.47] on held-out papers, at fixed team size and number of experienced authors (HELD-OUT SUPPORTED).

[1]Newcomer papers as a whole have 10.2 pp lower acceptance (−10.2 [−11.8, −8.7]) within topic and year (EXPLORATORY), but that gap mostly reflects who writes them: it shrinks once team size and the authors' ICLR experience are controlled, and shrinks further within the same authors.

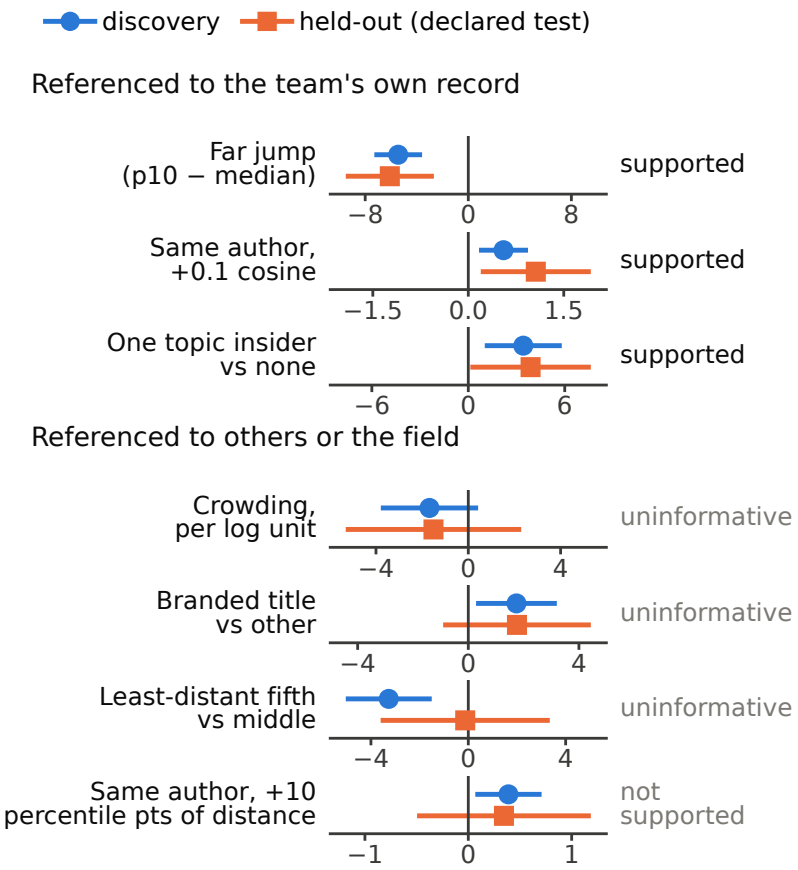


FIGURE 4.3: Gate tests by reference point: discovery estimates (blue circles) and declared held-out tests (orange squares), with 95% intervals. The held-out verdict is shown at right: 'supported' means HELD-OUT SUPPORTED; 'uninformative' means not supported and below the test's MDE80; 'not supported' marks a declared test without a registered MDE80, which is called inconclusive in the text. Each row has its own scale: pp for the far jump (closeness at its 10th percentile, p10, minus the median), the insider, the branded title and the least-distant fifth; pp per log unit for crowding; and pp per 0.1 cosine similarity and per 10 percentile points of distance for the two same-author slopes. Because the scales differ, compare verdicts rather than sizes.

The result does not survive the family-wise correction (Holm-adjusted $p = 0.052$), and status explains it equally well: each additional insider adds only 0.58 pp more than each additional experienced author from another topic ($p = 0.16$; EXPLORATORY).

Distance from earlier ICLR work as a whole gives uninformative results. In the same SPECTER2 vector space, papers in the least-distant fifth differ from those in the middle fifth by −3.27 [−4.93, −1.60] in discovery but by −0.13 [−3.50, +3.25] on held-out papers, a test with power = 0.47. The within-author slope for distance from earlier ICLR work was not supported either, and with no MDE80 available it is inconclusive. Of the distance-based tests, only the associations with the team's own earlier work held up on held-out papers.

Chapter 5

Conclusions and Future Work

This report set out to test whether research in machine learning herds. It separated two ways in which a field can conform, imitation of what others are doing and continuity with one's own record, and tested their predictions on ten years of ICLR submissions at three levels: the field as a whole, entry to a topic and the acceptance decision. This chapter first summarises the key findings (Section 5.1). It then discusses what they mean for each research question and how they relate to earlier work (Section 5.2), sets out the limitations (Section 5.3) and suggests future work (Section 5.4).

5.1 Summary of key findings

At the level of the field (RQ1), keyword waves are real. Of the 68 most frequent author keywords, 17 rose and then fell between 2019 and 2026 (EXPLORATORY). The number of topics that appear to be emerging, however, depends on how the topics are defined. The same papers give 8 emerging topics when the topic map is fitted on 2017–2020 and 29 when it is fitted on all years (EXPLORATORY). The one change that clearly lasted is the shift to foundation models: their share of submissions rose from 0.3% in 2017 to 54.4% in 2026, and topics in which more papers built on FMs in the previous year have fewer papers proposing a new architecture, by 8.0 pp in discovery and 11.0 pp on held-out papers for a change from no adoption to full adoption (REPLICATED). The registered tests of whether a topic's mix of contribution types changes as the topic grows were not supported.

At entry to a topic (RQ2), the registered herding prediction failed in the opposite direction. Papers by newcomers to a topic are more often method papers than papers by incumbents, by 3.7 pp in discovery and 3.6 pp on held-out papers, and less often measurement papers, by 1.7 pp and 2.1 pp (REPLICATED). The method surplus is linked to entering the topic rather than to being new to ICLR, since researchers who already had ICLR papers in other topics show it as well, and it appears in every period, including the years before 2023 (EXPLORATORY). Much of the measurement gap goes with team size, and in 2024–2026 measurement papers are concentrated among incumbents and larger teams. Newcomer papers also lie farther from earlier ICLR work, and first-time ICLR teams take up foundation models and young, rising keywords later than teams with an ICLR record (EXPLORATORY). Whether entrants flock to hot topics could not be decided (INCONCLUSIVE).

At the acceptance decision (RQ3), the registered test found no detectable penalty or premium for papers in crowded topics, but the test could not rule out effects smaller than about 5 pp per log unit (UNINFORMATIVE). Branded titles spread from 10.9% to 44.6% of titles and carry a small premium in discovery, which the held-out test was too weak to confirm (EXPLORATORY). The associations that held up on held-out papers all involve the team's own earlier work. Papers far from it are accepted 5.4 pp less often in discovery and 6.1 pp less often on held-out papers; for the same author, a paper closer to their earlier work is accepted more often; and a team with one topic insider is accepted more often than a team with none (all HELD-OUT SUPPORTED, although the insider result does not survive a family-wise correction). A topic that was favoured at review in 2018–2022 kept only part of its advantage in 2024–2026 (HELD-OUT SUPPORTED). FM papers are accepted more often, by 4.6 pp, and this premium did not shrink as FMs spread within a topic, while the timing of keywords shows at most weak associations with acceptance (both EXPLORATORY).

5.2 Discussion

5.2.1 Does machine learning herd?

Table 5.1 places each result under the route it fits. The field converged on foundation models, but imitation, continuity, lock-in and efficient choice all predict that, so the

	Entry (what is written)	Gate (acceptance)
Field, both routes: convergence on foundation models (REPLICATED), also predicted by lock-in and efficient choice.		
Imitation (others, now)	Follow-on prediction reversed (REPLICATED); newcomers farther from earlier work and fewer twins for returning topic newcomers (EXPLORATORY); flocking INCONCLUSIVE.	No detectable crowding premium (UNINFORMATIVE); branded titles: small premium, held-out test too weak (EXPLORATORY). Also fit a rival: topic edges partly kept (HELD-OUT SUPPORTED); no change in the FM premium; weak keyword timing (EXPLORATORY).
Continuity (a record)	Incumbents write more measurement papers (REPLICATED; much of it with team size); foundation models spread through teams with an ICLR record, as diffusion also predicts (EXPLORATORY).	Lower acceptance far from the team's record; higher for the same author's closer work and with a topic insider (HELD-OUT SUPPORTED; status and expertise fit too).

TABLE 5.1: Where conformity shows, by route and level. The table summarises Chapter 4 and adds no new estimates.

field level cannot settle the question. Where the two routes make different predictions, at entry and at the gate, the evidence leans towards continuity. The registered herding predictions for entry failed or were inconclusive, and the registered crowding test at the gate was uninformative. Every result that replicated at the gate is referenced to the team's own record, with one exception: the persistence of topic edges, which fits imitation of past acceptance and regression to the mean in a noisy review process equally well. The conformity that can be detected at ICLR is a pull towards a record rather than towards what others are doing.

This answer needs two qualifications. First, the absence of a detectable crowding premium does not rule out a bandwagon. Abrahamson and Rosenkopf [29] showed that bandwagons can spread even when adopting brings no gain, and the spread of branded titles may be such a case. Second, the record-side results at the gate fit continuity, but they do not single it out. The next subsections take each research question in turn.

5.2.2 The field: trends and hindsight

The hindsight result speaks to a common practice in the history of research fields. Studies of PNAS, computational linguistics and AI research have fitted one topic model to all years and then read trends from it [24, 25, 89, 90]. The present results show that, with the papers and the pipeline held fixed, the choice of fitting years alone can change

the count of emerging topics by a factor of more than three. This turns a concern raised in principle by Leydesdorff [95] and Rotolo et al. [26] into a measured difference, and it adds the time window to the list of analysis choices that a multiverse analysis would vary [102]. It also fits with recent work that tests signals of emergence using only earlier data [96]. The comparison is a bracket, not a correction: the early map cannot see topics that appeared after 2020, so neither count is the true number.

The shift to foundation models is the clearest case of convergence in the corpus, and it matches the accounts of paradigm shifts in NLP [86, 87] and the concern about homogenisation around shared models [4]. It is also what one would expect if general methods that scale with computation simply work better [85]. Because both routes and both of these readings predict it, the shift by itself says little about herding. This is the reason the report turned to entry and to the gate.

5.2.3 Entry: who brings what

The entry results run against the picture of the newcomer that underlies the herding account. In cascade models, the person with the least information follows those who chose before them [5, 6]. At ICLR, researchers new to a topic are the ones who more often propose methods, and their papers lie farther from earlier work. This fits the literature on outsiders, which finds that people from the margin of a field more often bring solutions and new ideas [62, 65, 66]. The present results add one detail that earlier work did not measure: the type of contribution. Newcomers bring methods, and incumbents and larger teams write more of the benchmarks and evaluations. This is consistent with the finding that ML relies on a small number of benchmark datasets [80], and with the observation in NLP that tasks and methods lead while datasets follow [88]. A method paper is not necessarily an original one, however, so these results say who proposes methods, not whose work is more novel.

First-time teams took up foundation models later than teams with earlier ICLR work. This fits absorptive capacity, the idea that new knowledge is taken up by those who already hold related knowledge [48]. It fits network diffusion equally well [35, 36], so it cannot separate the routes. It does, however, contradict the idea that the newest entrants rush to whatever is fashionable. Half of the first authors of LLM papers on

arXiv in 2023 were new to LLM research [22], but at ICLR, teams new to the venue did not lead the adoption of foundation models.

5.2.4 The gate: crowd or record

The strongest result at the gate is the far jump: papers far from the team's earlier work are accepted less often. This extends the pivot penalty, in which larger moves away from one's own past work go with lower impact and lower publication success [28], to the accept or reject decision of an ML venue under double-blind review, with every rejected paper in view. It also fits the finding that innovative research strategies are riskier [12]. The shape of the association matters: acceptance rises with closeness up to the fourth fifth of the distribution and then levels off, so leaving one's record is associated with lower acceptance, but staying very close brings no further gain. This reading suggests a threshold rather than a reward for repetition.

Several explanations fit these results, and the data cannot separate them. Continuity is one. Expertise is another: reviewers may judge work from a team that knows the area as stronger, and with reason, as grant reviewers are both biased towards and better informed about work in their own area [59]. Status is a third. Cumulative advantage favours those already recognised [60], famous authors fare better when reviewers can see them [78], and preprints can reveal authors under double-blind review [79]; at ICLR, a prestige gap has been tested with a pre-registered out-of-sample test [17]. The insider result is a good example. One insider on the team goes with higher acceptance, which matches the chaperone effect [69] and the finding that senior expertise offsets the disadvantage of newcomers [70]. But each additional insider adds little more than each additional experienced author from another topic, so a proven record in general may matter more than knowledge of the specific topic.

On the side of imitation, the gate shows little. The registered crowding test was too weak to detect effects of a few points, which is consistent with the high level of noise in ML review [72, 73]. Topic edges persist only in part, which noise also predicts, although differences between research areas in how reviewers score papers are a rival account [20]. Together with a randomised trial that found no herding among reviewers in discussions [43], the results give little support to the idea that ICLR review rewards following the crowd. Whether evaluators penalise novelty remains contested [14, 57], and the present

results do not settle it: the penalty found here is for distance from the team's own work, not for distance from the field, and the test of distance from earlier ICLR work as a whole was uninformative.

5.2.5 Why the distinction matters

The two routes carry different risks for a field. Imitation can discard private information and leave a consensus that rests on little evidence and breaks easily [5, 6]. Continuity trades long-run exploration for short-run competence [46], and choices that are safe for each researcher can make a whole field explore less [55]. A field that looks herded may in fact conform by continuity, and a diagnosis of fads would then point to the wrong problem. Acceptance is also only a short-run outcome. During the previous shift in ML, researchers who stayed close to their own record had lower later impact [23], so the work that is accepted and the work that matters later need not be the same. For those who write the history of ML and NLP, the results suggest a simple check: before reading trends from a topic model, refit it on early years and see which trends remain.

5.3 Limitations

All results are associations at one venue, and none measures what would change if a researcher chose a different topic, team, title or type of contribution. The held-out papers share conference years, reviewers and authors with the discovery papers. The held-out tests therefore guard against overfitting by the analyst, but not against patterns that are specific to a year or a pool of reviewers, and a supported held-out test confirms the direction of an association rather than its size. The standard errors may also be too small: under one simulated null, the main entry tests gave a significant result 12.6% and 10.0% of the time at a nominal 5% level.

The framework of the report was built partly after the analysis. The distinction between imitation and continuity was drawn up after the registered and declared tests had run, and the predictions of continuity were written afterwards. The record-side results at the gate were found post hoc. Two of the three rest on secondary held-out tests, and the insider result does not survive a family-wise correction. Keyword popularity and title style had been explored against acceptance before registration.

The measures have their own limits. Contribution and FM labels come from an LLM. They were consistent across two runs, but they were checked by hand only by the author, on 100 papers, about 20 of which were borderline, and there is no measure of agreement with a second human coder. A method paper is a type of contribution, not a measure of novelty, and distance in an embedding space is not originality. Entry status counts ICLR submissions only, and the change in author identifiers in 2021 hides earlier histories, which pulls the entry contrasts towards zero. Some abstracts and titles are final versions written after the decision; code links were removed for discovery papers, but titles could not be corrected. Withdrawn papers are left out of the acceptance tests. Finally, the hindsight comparison is a bracket, because the topic maps differ in the papers they are fitted on as well as in their time window.

5.4 Future work

The most direct next step is to separate status from continuity. If the record-side associations reflect status, they should be stronger where a preprint identifies the authors before review, which is how Ibrahim et al. [17] found prestige to operate. Running this test would require a new declaration before any outcome is read, as the protocol of this report requires.

The labels also need a human check that does not depend on one person. Two coders labelling the same random sample independently would give agreement between humans and between humans and the LLM, and would show whether the borderline cases change the entry results. A related question is what newcomers' method papers contain: whether they propose new ideas or adapt existing ones to a new topic.

Several open questions could be answered with more data. Whether entrants flock to hot topics remains unsettled, and a registered held-out test with enough power could settle it. Linking submissions to their later citations would show whether the work that is accepted more often is also the work that has lasting impact, as the persistence results of Bao and Li [23] suggest it may not be. Withdrawn papers and the text of reviews could show whether the gate also selects through withdrawal and what reviewers object to in papers that move far from their authors' earlier work. Data on compute

or institutions could show whether the concentration of measurement papers among incumbents reflects resources other than team size.

Finally, the design could be repeated at other ML and NLP venues, such as NeurIPS or ACL, to see whether the findings are specific to ICLR. Refitting topic maps on early years and using predictors measured only from the past could also become a routine check for claims about research trends.

5.5 Closing remarks

Machine learning is easy to describe as a field that chases fashions, and its keyword waves and the rapid spread of foundation models seem to confirm it. When the herding account is tested where it makes distinct predictions, however, it finds little support at ICLR. Newcomers to a topic bring methods rather than follow-on work, and acceptance shows no detectable association with following the crowd. What acceptance is associated with, instead, is staying close to the team's own earlier work. Whether that reflects expertise, status or the value of continuing a line of work is the main question this report leaves open.

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